

A Literature-Implied Positive Answer to Question 2.5 of arXiv:2102.03447: Model-Space Bessel Systems as a Vectorial Hankel Operator

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Status and source question

This is a `literature_implied_answer` (full positive answer) packet. Dayan asks in Question 2.5, on PDF page 8 of [1], whether

$$C := \sup_{z \in \mathbb{D}} \sum_{n \in \mathbb{N}} (1 - |\Theta_n(z)|^2) < \infty \quad (1)$$

forces the model spaces $K_{\Theta_n} = H^2 \ominus \Theta_n H^2$ to form a Bessel system, and asks separately whether this holds when all Θ_n are Blaschke products.

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Question 2.5. *Is any sequence of model spaces $(H_{\Theta_n})_{n \in \mathbb{N}}$ in H^2 satisfying*

$$(2.3) \quad \sup_{z \in \mathbb{D}} \sum_{n \in \mathbb{N}} 1 - |\Theta_n(z)|^2 < \infty$$

a Bessel system? Is it true if Θ_n is, for any positive integer n , a Blaschke product?

Remark 2.6. Question 2.5 has a positive answer whenever $\Theta_n = b_{\lambda_n}$ is, for any positive integer n , a degree-one Blaschke factor at a point λ_n , and therefore whenever H_{Θ_n} is the line spanned by the Szegő kernel at λ_n , [9] Ch. VI, Lemma 3.3].

The answer is yes for arbitrary inner functions. The supporting result is Treil's reproducing-kernel thesis for possibly vectorial pre-Hankel operators [2, Theorem 1.1]. Treil's theorem predates the source question and does not mention model spaces; the implication below is the direct operator-theoretic identification made in this run.

Identification

Let $P_- : L^2(\mathbb{T}) \rightarrow H_-^2 = L^2(\mathbb{T}) \ominus H^2$ denote projection onto the strictly negative Fourier frequencies, and put $E = \ell^2(\mathbb{N})$. For an inner function Θ ,

$$\Theta P_-(\overline{\Theta} f) = f - \Theta P_+(\overline{\Theta} f) = P_{K_\Theta} f, \quad f \in H^2. \quad (2)$$

Multiplication by Θ is isometric, so

$$\|P_-(\overline{\Theta}f)\|_2 = \|P_{K_\Theta}f\|_2. \quad (3)$$

The hypothesis at $z = 0$ makes the vectorial construction legitimate. In fact,

$$P_-\overline{\Theta}_n = \overline{\Theta}_n - \overline{\Theta}_n(0), \quad \|P_-\overline{\Theta}_n\|_2^2 = 1 - |\Theta_n(0)|^2,$$

and hence

$$\Phi := (P_-\overline{\Theta}_n)_{n \in \mathbb{N}} \in H_-^2(E).$$

It is therefore the antianalytic symbol of the vectorial pre-Hankel operator

$$\Gamma f = P_-(\Phi f) = (P_-(\overline{\Theta}_n f))_{n \in \mathbb{N}}, \quad (4)$$

initially defined on analytic polynomials. The second equality holds because subtracting the constant $\overline{\Theta}_n(0)$ does not affect P_- after multiplication by an analytic f .

Full answer

Theorem 1. *Let $(\Theta_n)_{n \in \mathbb{N}}$ be any sequence of inner functions satisfying (1). Then $(K_{\Theta_n})_{n \in \mathbb{N}}$ is a Bessel system, and one may take Bessel bound $4eC$:*

$$\sum_{n \in \mathbb{N}} \|P_{K_{\Theta_n}}f\|_2^2 \leq 4eC \|f\|_2^2, \quad f \in H^2.$$

Conversely, a Bessel bound B implies (1) with $C \leq B$. Thus the kernel condition characterizes Bessel systems of model spaces.

Proof. Let

$$k_z(w) = \frac{(1 - |z|^2)^{1/2}}{1 - \bar{z}w}$$

be the normalized Szegő kernel. For an inner Θ , the adjoint multiplier identity $M_\Theta^* k_z = \overline{\Theta(z)} k_z$ gives

$$\|P_{K_\Theta} k_z\|_2^2 = \langle P_{K_\Theta} k_z, k_z \rangle = 1 - |\Theta(z)|^2. \quad (5)$$

Combining (3), (4), and (5),

$$\|\Gamma k_z\|_{H_-^2(E)}^2 = \sum_n \|P_{K_{\Theta_n}} k_z\|_2^2 = \sum_n (1 - |\Theta_n(z)|^2) \leq C.$$

Treil's theorem applies because the domain is scalar H^2 , the target is the vector-valued space $H_-^2(E)$, and Γ is a vectorial pre-Hankel operator. It yields

$$\|\Gamma\| \leq 2\sqrt{e} \sqrt{C}.$$

For analytic polynomials f , equations (3) and (4) now give

$$\sum_n \|P_{K_{\Theta_n}}f\|_2^2 = \|\Gamma f\|_{H_-^2(E)}^2 \leq 4eC \|f\|_2^2.$$

Density extends the inequality to every $f \in H^2$. Conversely, applying a Bessel inequality to the unit vector k_z and using (5) gives the displayed kernel condition, uniformly in z . This proves both assertions. $\square$

$$\|k_\lambda\|_2 = 1 \quad \text{and} \quad \|k_\lambda\|_2 = 1 - \overline{\lambda}z, \quad \lambda \in \mathbb{D}.$$

and that $\|k_\lambda\|_2 = 1$.

The goal of this paper is to give an elementary proof of the following well known result.

Theorem 1.1 (Reproducing kernel thesis for Hankel operators). *Let Γ be a possibly vectorial (pre)Hankel operator such that*

$$\sup_{\lambda \in \mathbb{D}} \|\Gamma k_\lambda\|_2 \leq A < \infty$$

Then Γ is bounded and $\|\Gamma\| \leq 2\sqrt{e}A$.

This theorem for the scalar-valued case (with some constant C instead of $2\sqrt{e}$) was published in [1], and is widely used in theory of Hankel operators.

The proof presented in this note is quite elementary and uses only Green’s formula: the standard proof uses Nehari Theorem, H^1 -BMO duality and the fact that the so-called Garsia norm is an equivalent norm in BMO.

Provenance, search bounds, and limitations

Treil’s arXiv revision explicitly says that its proof covers vectorial Hankel operators, and Theorem 1.1 states the needed scalar-domain/vector-target result with the constant $2\sqrt{e}$. Dayan’s bibliography does not cite Treil, and the supporting paper predates and does not explicitly discuss Dayan’s question. Accordingly this packet records an agent-identified literature implication, not a newly proved analytic theorem.

The run’s cheap indexes were searched for the source arXiv id, title, and core keywords. A bounded web/arXiv search used the exact displayed condition and question wording, together with “model-space Bessel system,” “vectorial Hankel operator,” and “reproducing kernel thesis.” No later paper explicitly answering Question 2.5 was found; the reformulation led to [2].

There is no Blaschke-product limitation: the proof covers every countable sequence of inner functions. The inherited constant $4e$ is explicit but is not asserted to be optimal. Human review should focus on (2) and the vectorial-symbol construction; these are the only identifications beyond the cited theorem.

References

- [1] A. Dayan, *Weakly Separated Bessel Systems of Model Spaces*, arXiv:2102.03447v2 (2021), Question 2.5, arXiv:2102.03447.
- [2] S. Treil, *A remark on the reproducing kernel thesis for Hankel operators*, arXiv:1201.0063v2 (2012), Theorem 1.1; St. Petersburg Math. J. **26** (2015), 479–485, arXiv:1201.0063.